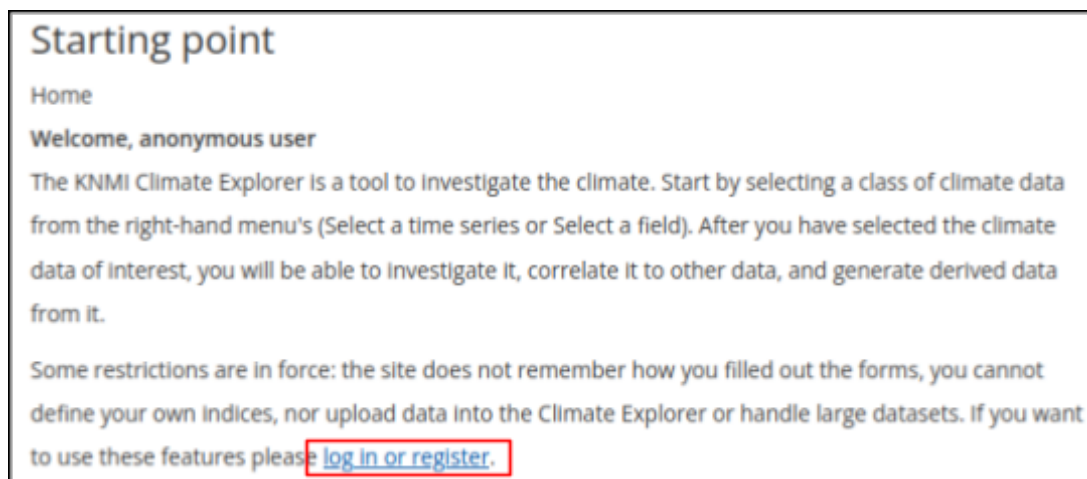


Downloading gridded data from the Climate Explorer

KNMI's Climate Explorer tool provides easy access to a lot of different datasets.

To find them, go to <https://climexp.knmi.nl/select.cgi>

On the landing page we're prompted to log in. This will unlock a few features, including being able to retrieve data queries that you've run within the last three days.



Starting point

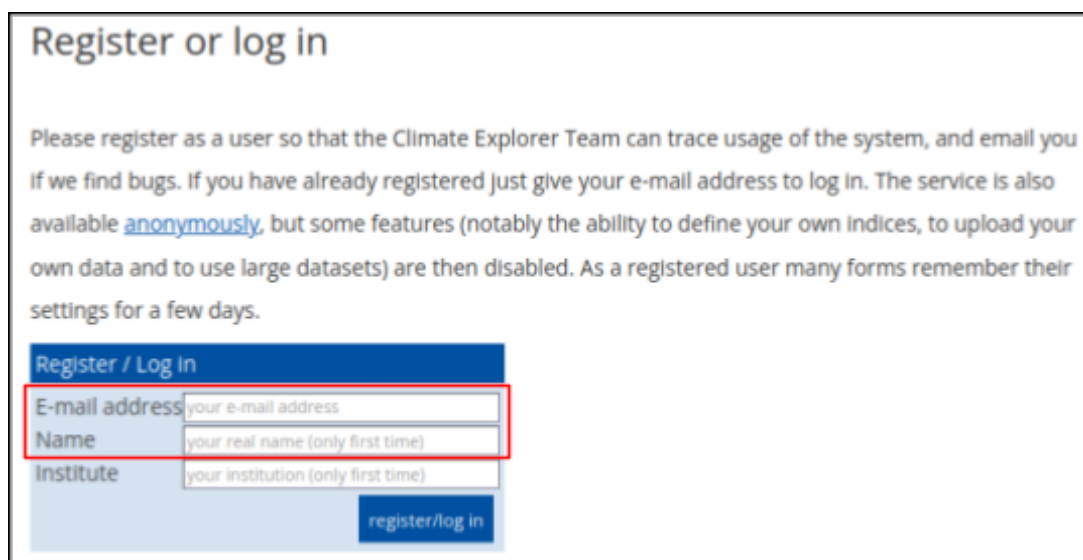
Home

Welcome, anonymous user

The KNMI Climate Explorer is a tool to investigate the climate. Start by selecting a class of climate data from the right-hand menu's (Select a time series or Select a field). After you have selected the climate data of interest, you will be able to investigate it, correlate it to other data, and generate derived data from it.

Some restrictions are in force: the site does not remember how you filled out the forms, you cannot define your own indices, nor upload data into the Climate Explorer or handle large datasets. If you want to use these features please [log in or register](#).

You need to provide an email address and (on your first visit) your name. This is just used to generate a unique user ID to retrieve your queries later.



Register or log in

Please register as a user so that the Climate Explorer Team can trace usage of the system, and email you if we find bugs. If you have already registered just give your e-mail address to log in. The service is also available [anonymously](#), but some features (notably the ability to define your own indices, to upload your own data and to use large datasets) are then disabled. As a registered user many forms remember their settings for a few days.

Register / Log in	
E-mail address	your e-mail address
Name	your real name (only first time)
Institute	your institution (only first time)
register/log in	

On the menu on the right-hand side of the screen, scroll down to 'daily fields' - this is where we'll get our gridded daily data from.



The top section contains gridded data products derived from **observations**, usually with some sort of modelled interpolation to fill in the gaps.

The subsequent sections contain **reanalysis** data. These use a weather forecasting model to assimilate a large amount of weather variables and predict (or 'hindcast') historical weather.

Scroll down and select the dataset and weather variable you're looking for. Different datasets provide different variables and cover different regions and time periods. Note that ERA5 data is provided at 0.25° resolution for Europe, Africa and North America, and globally at 0.5° resolution.

You'll be taken to a landing page with metadata for the selected dataset - you can check that you've picked a dataset that covers the region and time period you're interested in.

Field

ERA5 T2m

"ERA5 reanalysis, <https://www.ecmwf.int/en/forecasts/datasets/reanalysis-datasets/era5>" averaged over 2x 2 grid boxes Spatial coverage

X axis: whole world in 720 0.50° steps, first point at 0.12° E, last point at 359.62° E

Y axis: regular grid with 360 -0.50° steps, first point at 89.88° N, last point at 89.62° S

1-daily data available from 01Jan1950 to 31May2026 (27968 times) Temporal coverage

Variable t2m (Near-Surface Air Temperature) in K

[Full metadata](#). The associated [land/sea mask](#) is available for some operations

For ERA5, the reanalysis data only extends to the end of the last calendar month processed (usually with a latency of a week or so). The Climate Explorer also provides an extended version of the dataset, which includes analysis (operational reanalysis for the current month that hasn't been fully validated yet) and forecast data for the last week. To access it, scroll down to the bottom of the page and click the 'extended version' link.

[Analyse ERA5 T2m extended with operational analyses](#)

(Please contact [the administrator](#) if you need an up-to-date version)

[extended version](#)

Download ERA5 T2m

Contains modified Copernicus Climate Change Service information. Please read and agree to the [Copernicus license](#) before downloading.

Please consider downloading this field from the [authoritative site](#).

If you *really* want to get it here, ERA5 T2m is available as a netcdf file: (size 21098.2 MB).

The landing page will now show the extended date range:

Field

ERA5 T2m

"ERA5 reanalysis, <https://www.ecmwf.int/en/forecasts/datasets/reanalysis-datasets/era5>" averaged over 2x 2 grid boxes

X axis: whole world in 720 0.50° steps, first point at 0.12° E, last point at 359.62° E

Y axis: regular grid with 360 -0.50° steps, first point at 89.88° N, last point at 89.62° S

1-daily data available from 01Jan1950 to **15Aug2026** (28044 times)

Variable t2m (Near-Surface Air Temperature) in K

[Full metadata](#). The associated [land/sea mask](#) is available for some operations

To extract of a subset of the gridded data (whether a reanalysis or observational dataset), you need to define the region to subset, and also (very important!) confirm that you want to subset the data and not average over the box.

Get grid points, average area or generate subset

Mask: [add a mask to the list](#)

Latitude: °N - °N (ignored if mask selected)

Longitude: °E - °E **Define spatial extent**

Year range: -

Boundaries:

Make: ☐ average ☐ max ☐ min ☐ set of grid points ☒ **subset of the field** **Subset the data**

Considering: ☒ everything ☐ land points ☐ sea points [show/hide more](#)

Units: ☒ **convert to Celsius** ☐ leave in K **Convert to your preferred units**

[Create](#)

When you've made your choices, press the 'create' button to process them.

This can take quite a while...

... but when it's finished, scroll down to the bottom of the page and press 'download' to be taken to a page with a download link.

Download ERA5_-11-36E_33-66N_firstyear-lastyear T2m

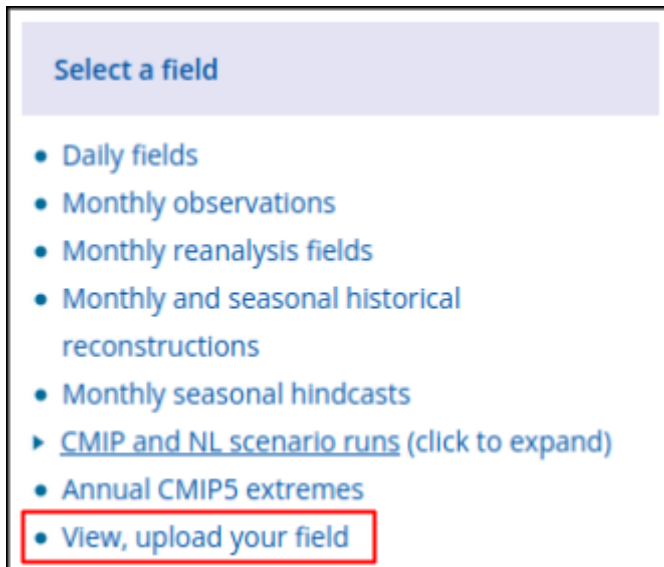
Please consider downloading this field from the [authoritative site](#).

If you *really* want to get it here, ERA5_-11-36E_33-66N_firstyear-lastyear T2m is available as a netcdf file: (size 1935.69 MB).

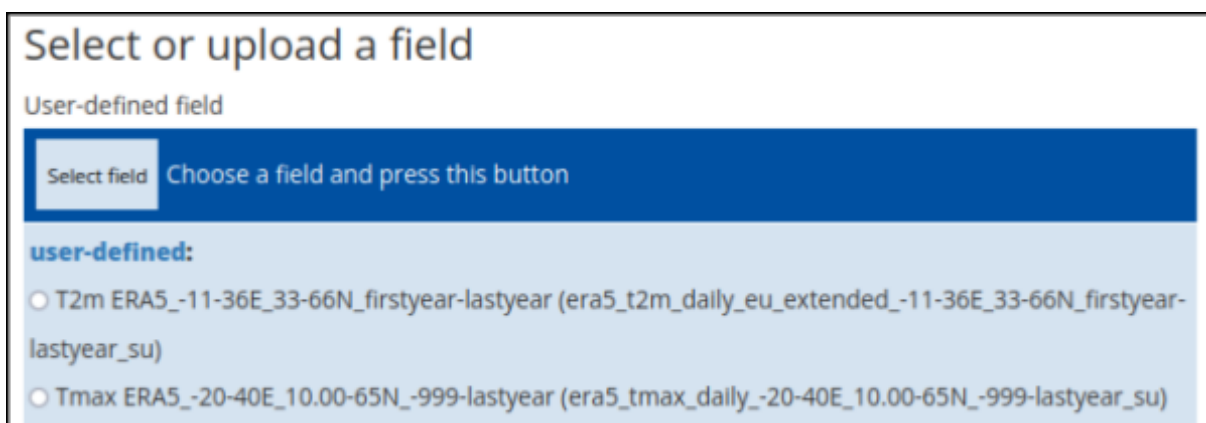
[Download](#)

Returning to your query later

If you want to revisit a file you create within the last three days, go back to the right-hand menu and select 'view, upload your field'



You'll be shown a list of available files - just select the one you want and press 'select', and you'll be taken back to the subsetted data.



Useful datasets

ERA5 - Global reanalysis from 1950 to near-real-time, produced at $\sim 0.25^\circ$ resolution. Provides most weather variables. More variables are available at the [Copernicus Climate Data Store](#).

E-Obs - Europe-only observations-based gridded data product from 1920 to end of previous calendar month, produced at 0.1° or 0.25° resolution. Station coverage has changed over the years so should be checked carefully. Provides temperatures and precip.

CHIRPS - Global precipitation data at 0.25° resolution between 50S and 50N, from 1981 to end of previous calendar month. Derived from satellite data.

CPC - Global temperature and precipitation data at 0.5° resolution, from 1979 to near-real time. Derived from station data so may not perform well in sparse regions.